Staff Materials Engineer

MetOx International · Houston, TX

Job ID (this archive)	ENERGY-023
Employer	MetOx International
Location(s)	Houston, TX
Career stage (archive tag)	Experienced (Senior)
Location group	Texas (outside DFW)
Majors (archive tags)	Materials Science & Engineering
Energy area	Advanced materials for energy
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 25, 2026.
Source page	https://job-boards.greenhouse.io/metoxinternationalinc/jobs/5374957008
Original link (from the course sheet)	https://to.indeed.com/aamtfxfnlht8
Captured	2026-10-25
Course sheet refs	ENERGY-023

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Job description

Empower Your Future: At MetOx International, we're pioneering the next era of energy security

and abundance through breakthrough superconducting technology. As a Staff Materials Engineer focused on high-temperature superconducting (HTS) materials, you'll join a multidisciplinary team advancing the materials science that enables next-generation superconducting wire. Your work will directly influence superconducting performance, manufacturing capability, and the commercialization of one of the world's most transformative energy technologies.

Based at our global headquarters in Houston, TX, the Staff Materials Engineer provides technical leadership in the development, characterization, optimization, and scale-up of high-temperature superconducting materials and related complex oxide systems that support MetOx's manufacturing platform. Working closely with Process Engineering, Manufacturing Engineering, Equipment Engineering, Quality, and R&D, this role combines advanced materials science, thin-film engineering, and process development to establish technical direction, improve manufacturing robustness, and accelerate the commercialization of next-generation superconducting technologies.

Key Responsibilities

Materials Development Strategy

- Leads the materials strategy supporting MetOx's HTS manufacturing platform and long-term technology roadmap.
- Develops a first-principles understanding of the relationships between MOCVD process conditions, film growth, microstructure, and superconducting performance.
- Defines microstructural targets (texture, defects, interfaces, pinning centers, and composition) required to maximize in-field critical current and manufacturing robustness.
- Identifies and evaluates new precursor chemistries, deposition approaches, and characterization methods to improve performance, yield, and scalability.
- Drives the transition of laboratory discoveries into robust, high-volume manufacturing processes.

Project & Experiment Leadership

- Designs and leads experimental programs to establish processing-structure-property relationships in HTS thin films.

- Develops MOCVD process windows through systematic experimentation, physics-based modeling, and statistical analysis.
- Applies advanced characterization techniques, including XRD, SEM, TEM, EBSD, AFM, Raman spectroscopy, and XPS, to identify performance-limiting mechanisms.
- Leverages large manufacturing datasets using multivariate statistics, machine learning, and artificial intelligence to correlate tool parameters, diagnostics, and film performance.
- Develops new in situ and ex situ diagnostic methods to improve understanding and control of film growth.

Integration and Cross-Functional Leadership

- Partners with Process, Equipment, Manufacturing, Quality, and Operations teams to transition new materials and processes into production.
- Translates scientific understanding into manufacturable process improvements that increase yield, consistency, and performance.
- Collaborates with suppliers, universities, national laboratories, and strategic partners to accelerate technology development.

Mentorship and Technical Leadership

- Provides technical leadership and mentorship to engineers and scientists.
- Promotes rigorous experimental design, data-driven decision-making, and scientific excellence.
- Serves as the subject matter expert for HTS materials, MOCVD process science, and superconducting performance.
- Guides technical problem solving and supports critical engineering decisions across research, process development, and manufacturing teams.
- Builds technical capability by coaching engineers, fostering collaboration, and cultivating a culture of continuous learning and innovation.

Minimum Qualifications

- Master’s in Materials Science, Physics, Chemical Engineering, Chemistry, or a related discipline. An advanced degree may be substituted for required experience on a year-for-year basis, depending on relevance and level.
- 7+ years of experience developing HTS materials, complex oxides, epitaxial thin films, or closely related functional materials.
- Deep understanding of processing–structure–property relationships in advanced materials.
- Experience with thin-film growth, epitaxy, crystal defects, microstructure evolution, and advanced materials characterization.
- Demonstrated ability to design and execute complex experimental programs.
- Experience applying multivariate statistics, machine learning, or AI to large experimental or manufacturing datasets.
- Strong scientific foundation with the ability to connect first-principles materials behavior to manufacturing outcomes.

Preferred Qualifications

- Ph.D. in Materials Science and Engineering, Materials Engineering, Physics, Chemistry, Chemical Engineering, or a related engineering or scientific discipline.
- Direct experience with REBCO, YBCO, or related high-temperature superconductors.
- Expertise in MOCVD or other epitaxial thin-film deposition methods (PLD, sputtering, ALD, CVD).
- Experience developing vortex pinning architectures and optimizing in-field critical current.
- Experience transferring research into pilot-scale or manufacturing environments.
- Record of technical leadership demonstrated through publications, patents, or commercialization of advanced materials technologies.

Physical Demands: This position requires sitting at a desk and viewing a computer screen for

extended periods while working or performing other office tasks. Ability to move around the office, including occasional standing, walking, and bending to retrieve documents, attend meetings, or use office equipment. Some light lifting may be required, up to 20 lbs.

MetOx is proud to offer competitive benefits including:

- Health, dental, and vision available on the first day of the month following your start date
- 401(k) match
- Paid parental leave & adoption assistance
- Educational reimbursement
- And more!

We are an equal opportunity employer and all qualified applicants will receive consideration for employment without regard to race, color, religion, sex, national origin, disability status, protected veteran status, or any other characteristic protected by law.

Archive notes for ECS 1100

Retrieval note: Official MetOx requisition 5374957008 matches title and Houston; page includes a live application form.

Key skills listed in the archive: superconductors; materials characterization; processing; failure analysis; R&D

Archive metadata — Accessed October 25, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer / ATS

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Identity review: Title, company and location matched (see Notes). Source type: Official employer / ATS. Source identifier: 5374957008.

Sources and retrieval details

Original workbook URL

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